

In the study of human neural networks, when observing neurons and the connections between them, the characteristics of a small world network can also be identified.

It is also known that there are more primitive living organisms whose brains are less convoluted and contain fewer neurons and connections.

It would be interesting to examine this aspect of evolutionary development, studying species that evolved from one another and possess increasingly complex brains. The focus of such an investigation would be to map their neural networks, which could shed light on the evolutionary origins of small-world networks: how they appeared in nature and why they became part of evolution itself.

What is the correlation or natural phenomenon that has created human social networks similar in structure to our neural networks? Could the small-world structure of our social relationships be a reflection, result, or consequence of our neural system?

We need to understand brain evolution better, and if we can simulate the current human neural network evolution process, we may improve these structures to find neural networks for deep learning which may be a significant step forward to general artificial intelligence, as most probably it would think in a similar way as humans. So stakeholders are AI researchers and programmers.

research paper ideas:

<https://arxiv.org/pdf/1904.04862>

<https://www.sciencedirect.com/science/article/abs/pii/S0378437116302849>